



DIGITAL VLSI DESIGN

Prof. Mahesh Awati & Prof. Vinay Reddy Narayana
Department of Electronics and Communication Engg.

DIGITAL VLSI DESIGN

PART 2

Fabrication of MOSFETs with Layout Design

Vinay Reddy Narayana

Department of Electronics and Communication Engineering

Fabrication of MOSFETS with Layout Design

UNIT 1 (Part 2): Fabrication of MOSFETS with Layout Design

Introduction, Fabrication Process Flow: Basic Steps (2.1 & 2.2)

The CMOS n-Well Process (2.3)

Layout Design Rules (2.4)

Full-Custom Mask Layout Design (2.5)

DIGITAL VLSI DESIGN

Fabrication of MOSFETs

Part b – CMOS Fabrication Process

Mahesh Awati & Vinay Reddy Narayana

Department of Electronics and Communication Engineering

Fabrication of MOSFETs & Circuit Design Process

CMOS Fabrication Process

Formation of P-Substrate by adding P Type Impurity



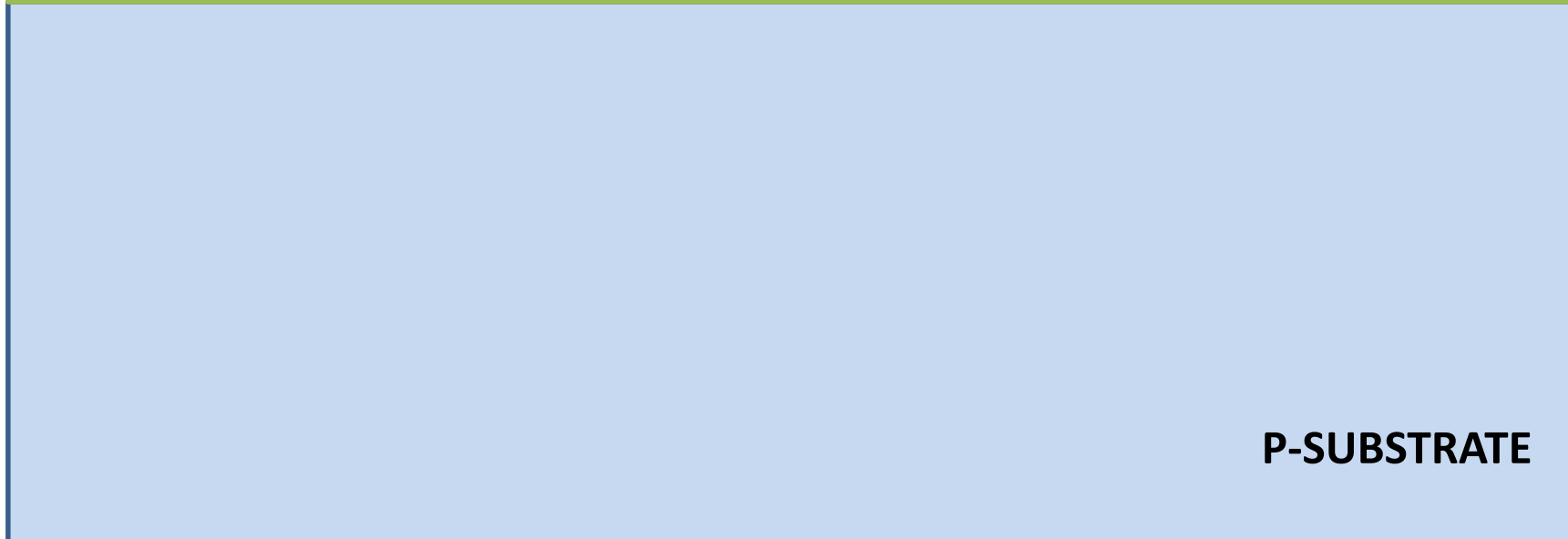
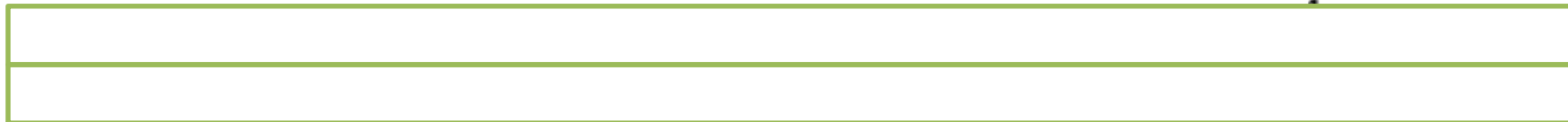
A large, solid light-blue rectangle represents the P-substrate. It is centered within a larger white area that is enclosed by an orange border. The text 'P-SUBSTRATE' is located in the bottom right corner of this rectangle.

P-SUBSTRATE

Fabrication of MOSFETs & Circuit Design Process

CMOS Fabrication Process

Formation of Active Region for N-well



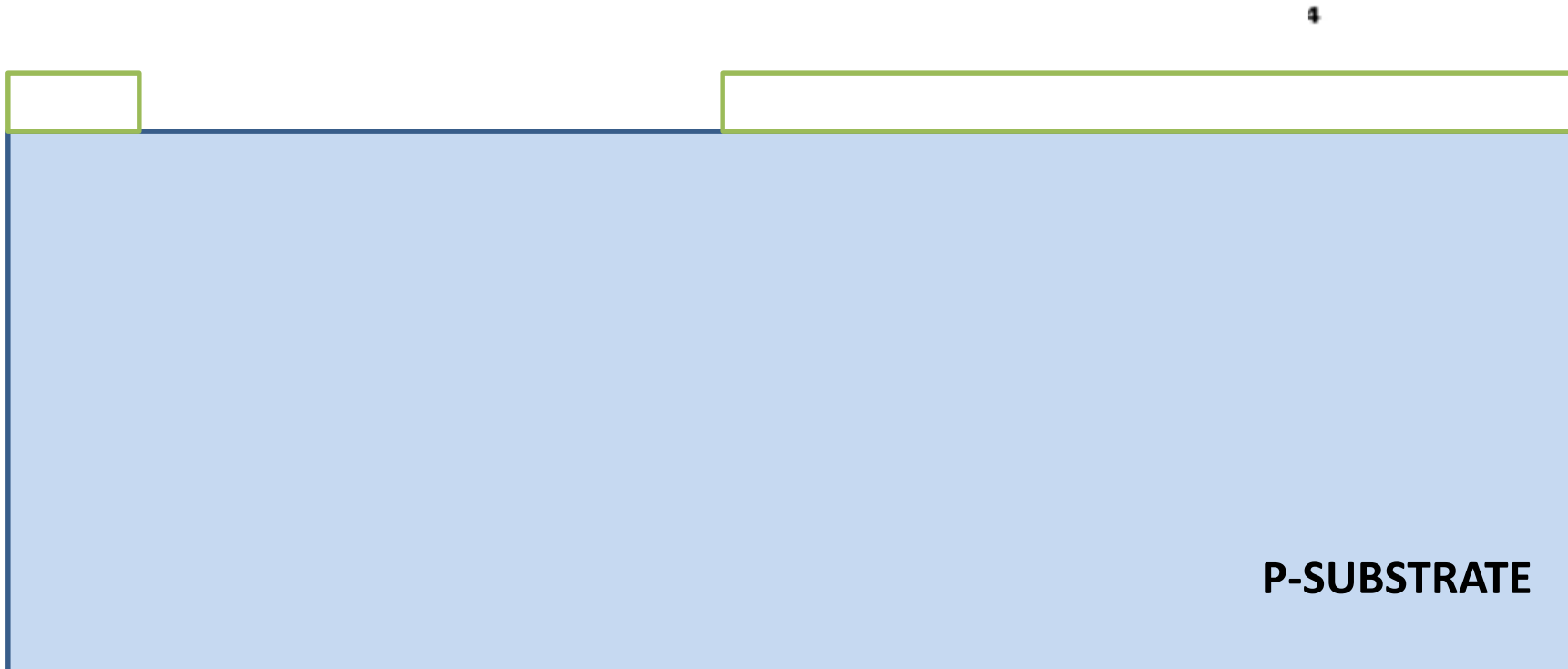
P-SUBSTRATE

1. SiO₂ over the Surface
2. Photoresist
3. Photolithography MASK1
4. Etching

Fabrication of MOSFETs & Circuit Design Process

CMOS Fabrication Process

Diffusion of N-well in the Active Region where PMOS device to be fabricated



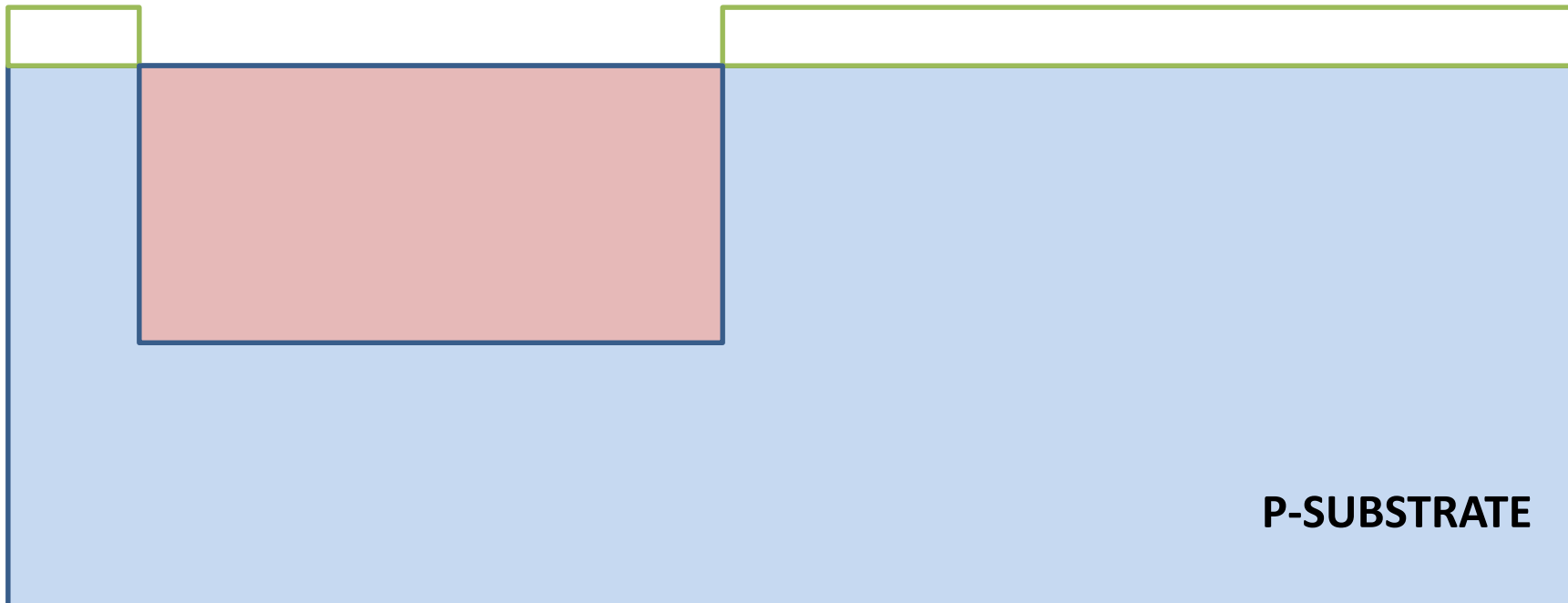
Fabrication of MOSFETs & Circuit Design Process

CMOS Fabrication Process

Formation of Active Region for nmos Device



4



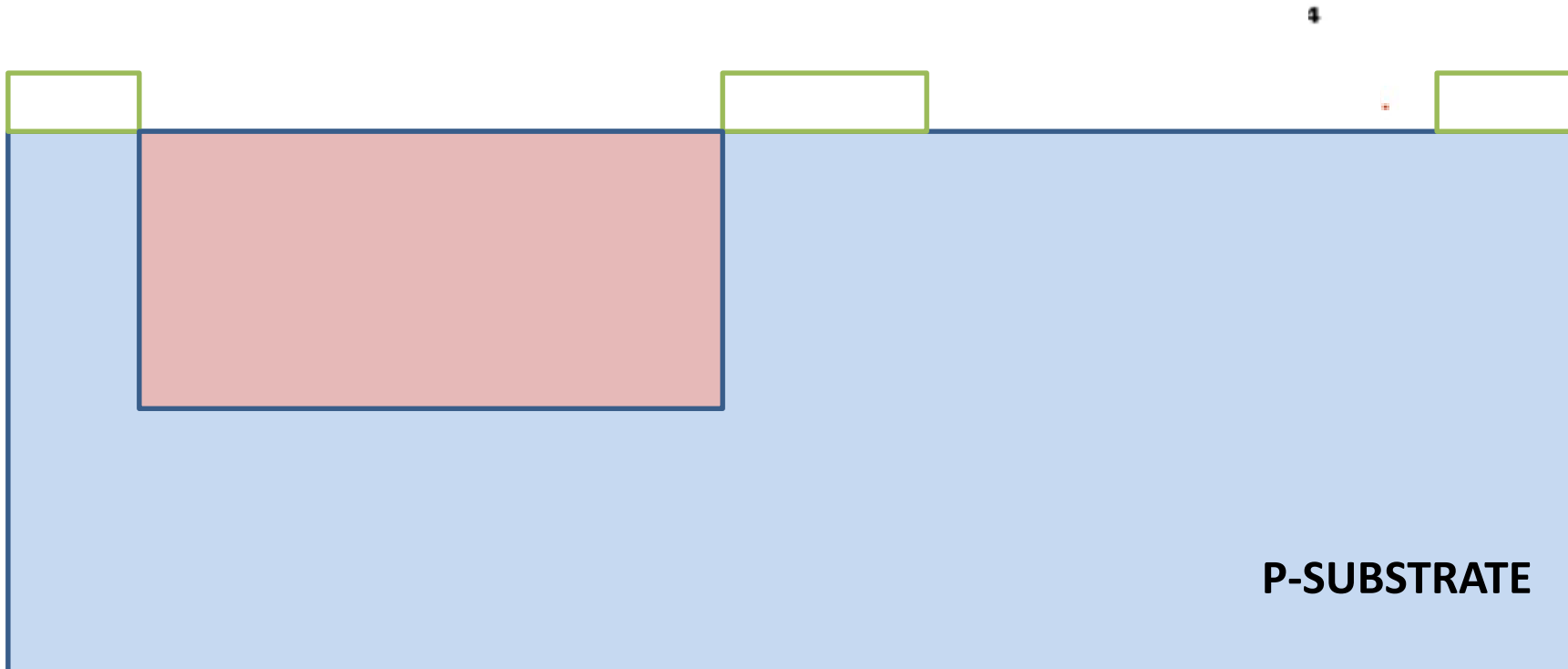
P-SUBSTRATE

1. SiO₂ over the Surface
2. Photoresist
3. Photolithography MASK2
4. Etching

Fabrication of MOSFETs & Circuit Design Process

CMOS Fabrication Process

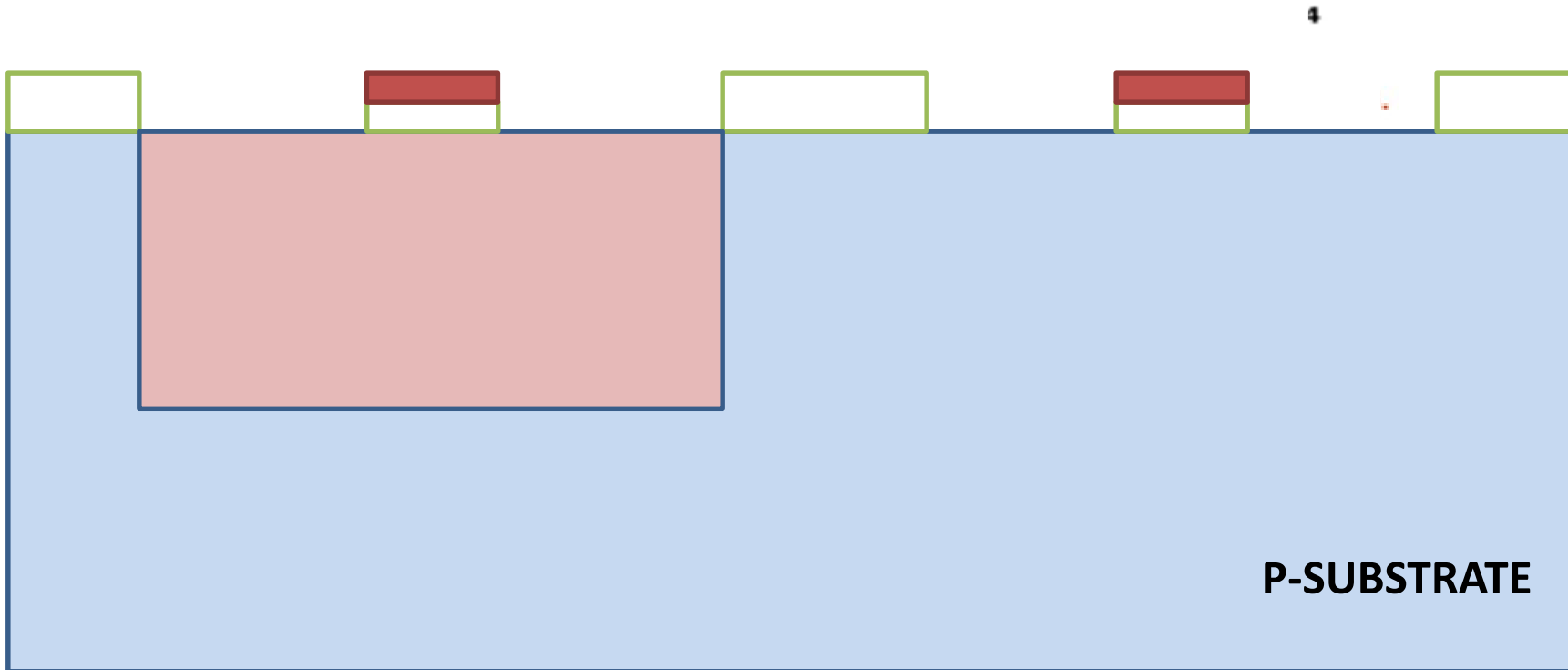
Formation of GATE for NMOS and PMOS



Fabrication of MOSFETs & Circuit Design Process

CMOS Fabrication Process

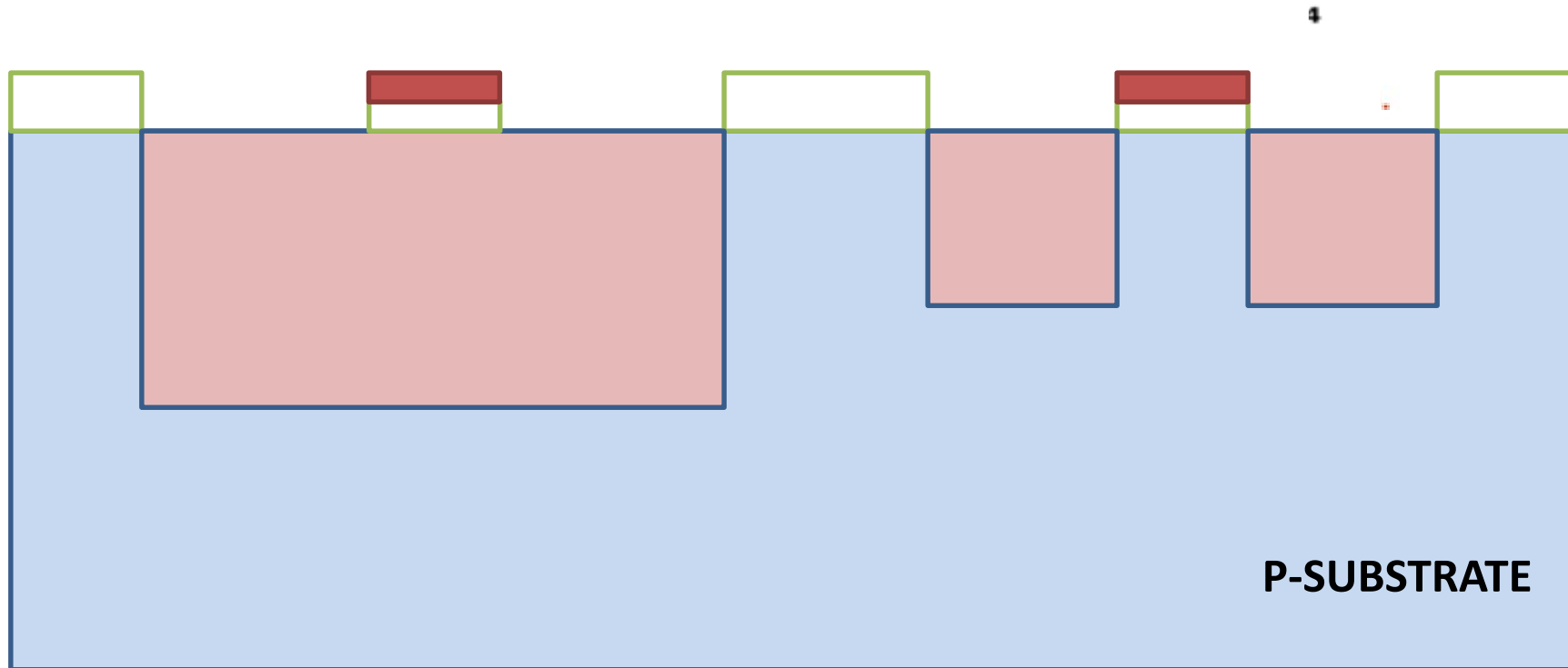
Diffusions of n+ to form NMOS



Fabrication of MOSFETs & Circuit Design Process

CMOS Fabrication Process

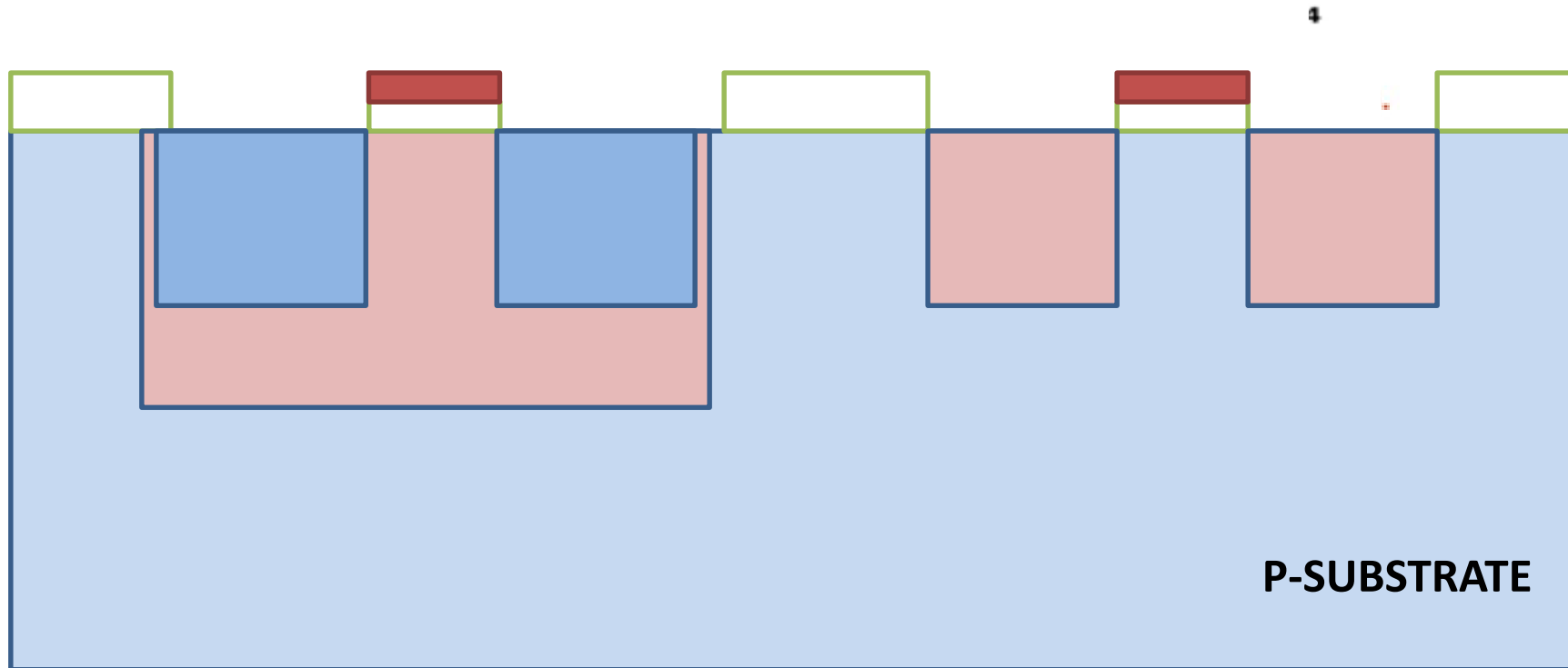
Diffusions of P+ to form NMOS



Fabrication of MOSFETs & Circuit Design Process

CMOS Fabrication Process

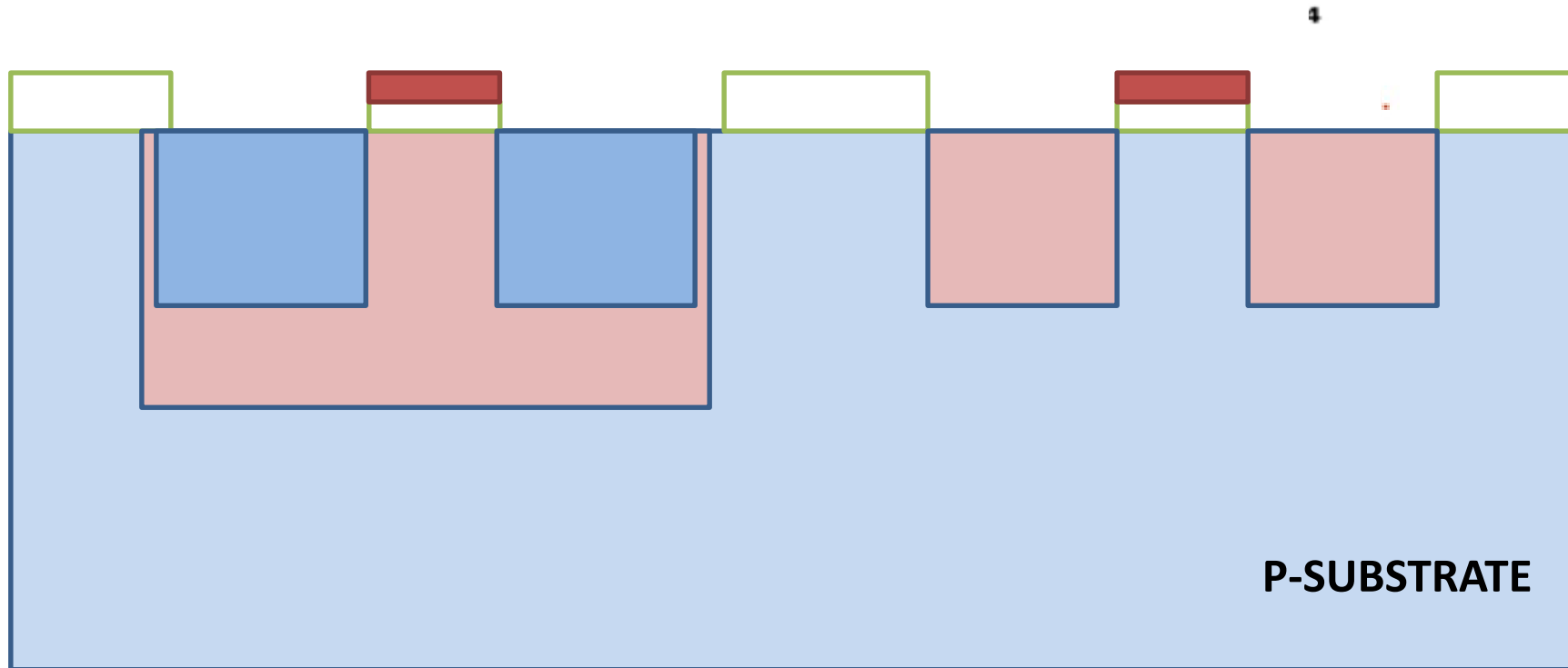
Diffusions of P+ to form NMOS



Fabrication of MOSFETs & Circuit Design Process

CMOS Fabrication Process

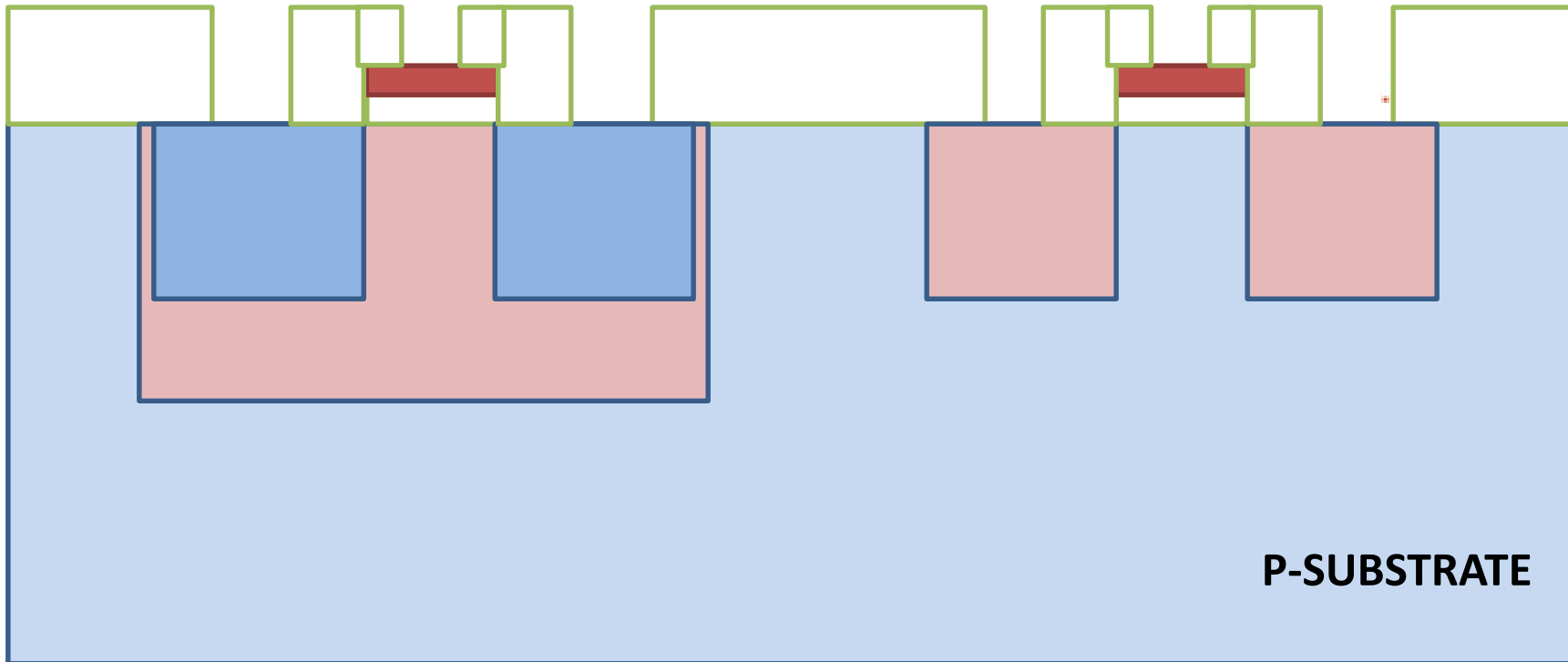
Formation of Contact cut



Fabrication of MOSFETs & Circuit Design Process

CMOS Fabrication Process

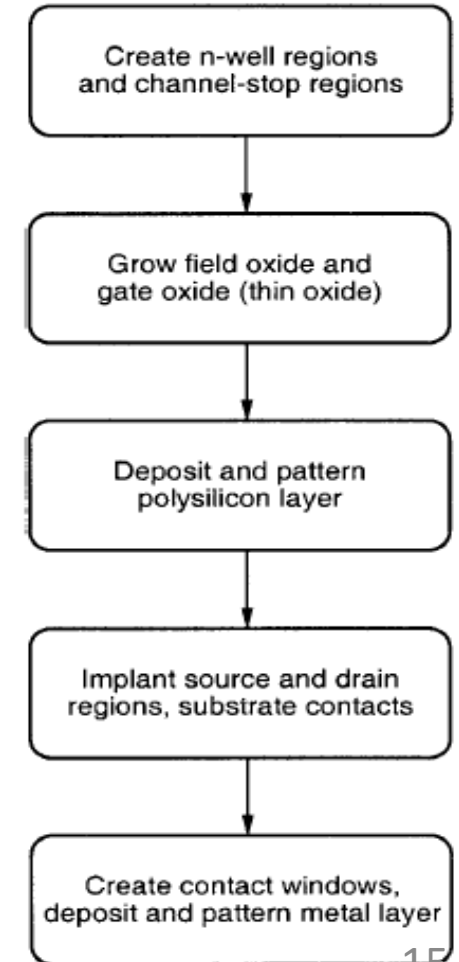
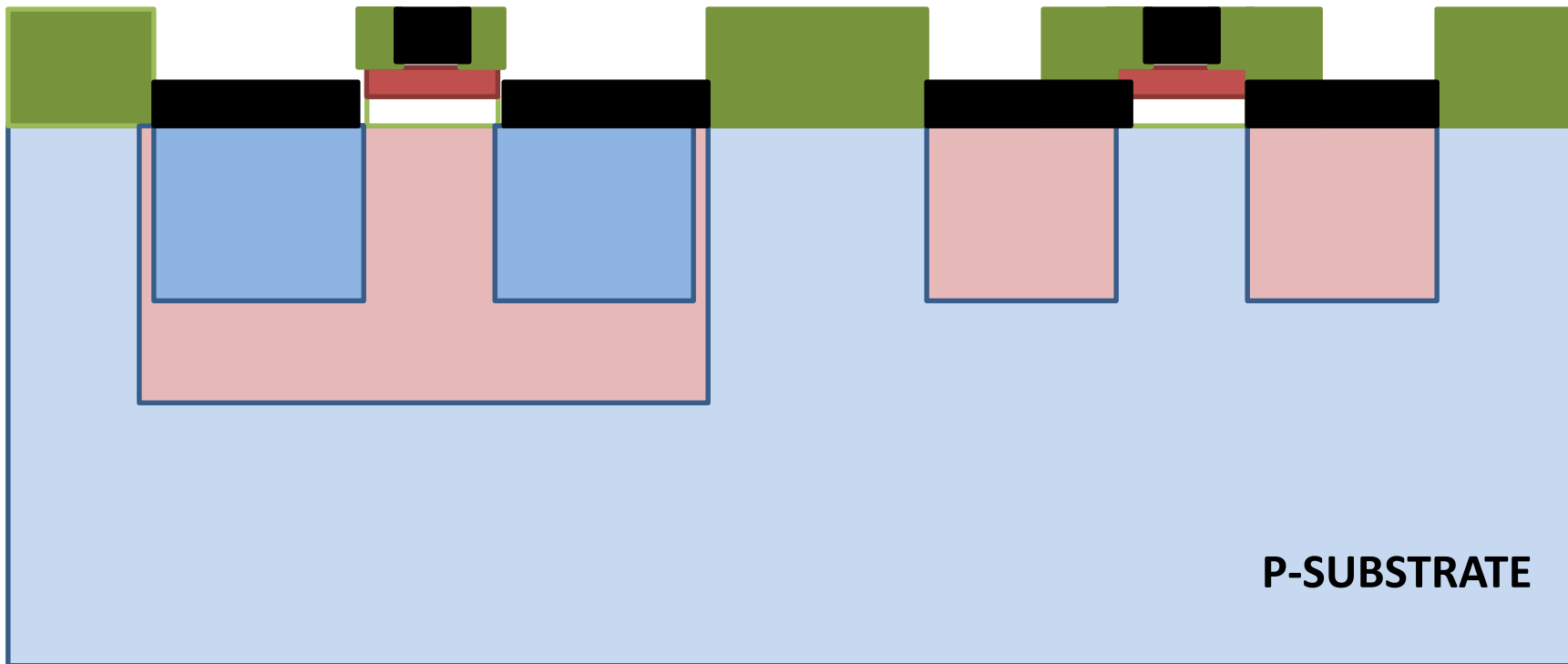
Formation of Contact cut



Fabrication of MOSFETs & Circuit Design Process

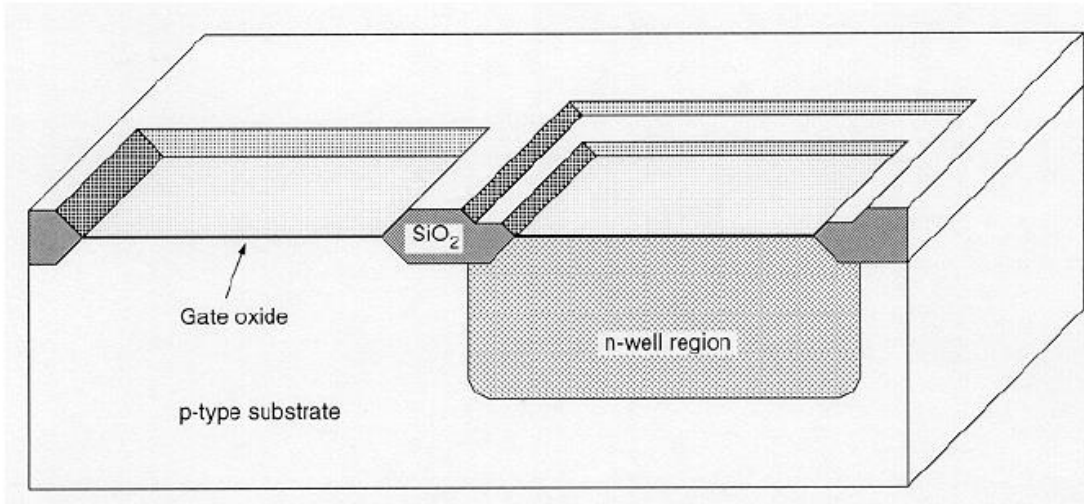
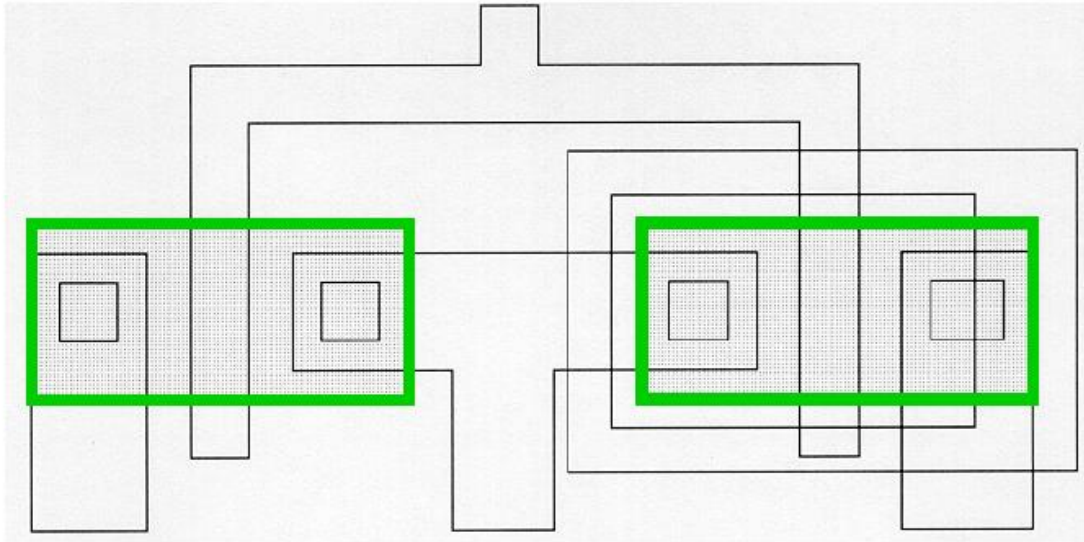
CMOS Fabrication Process

Formation of Contact cut



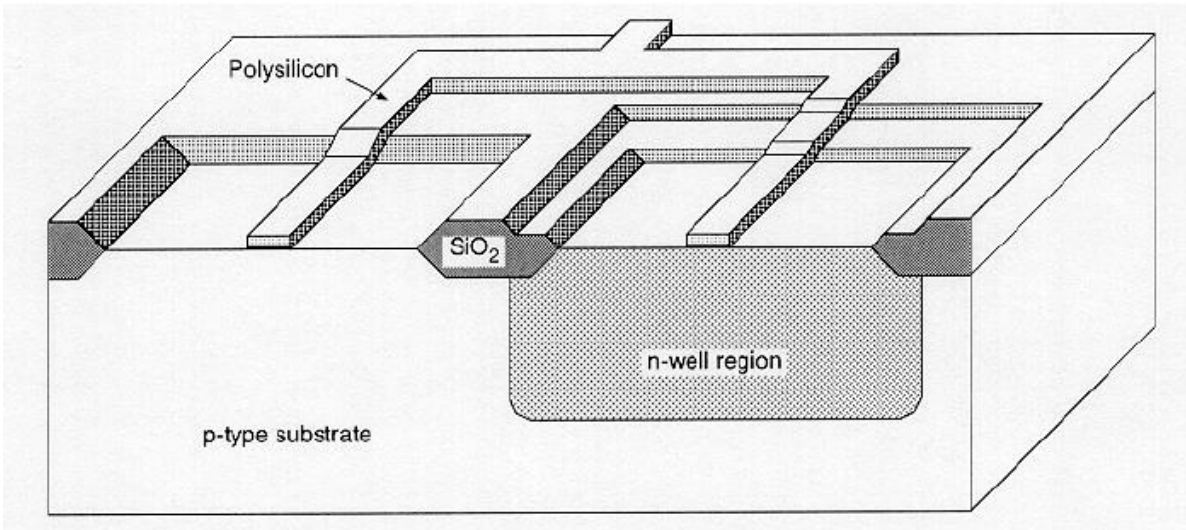
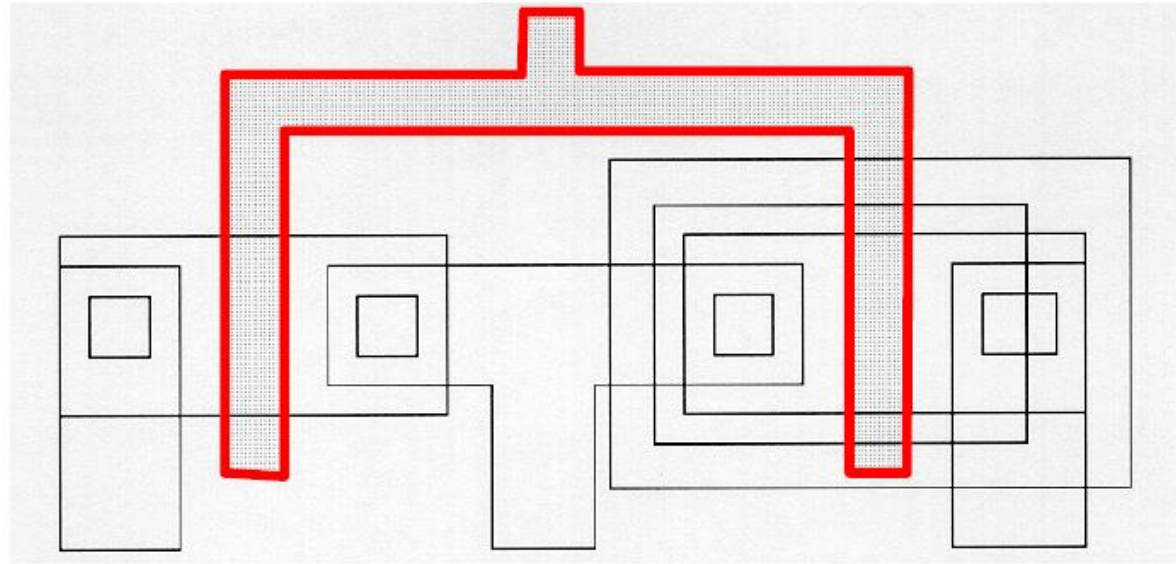
CMOS Process

active



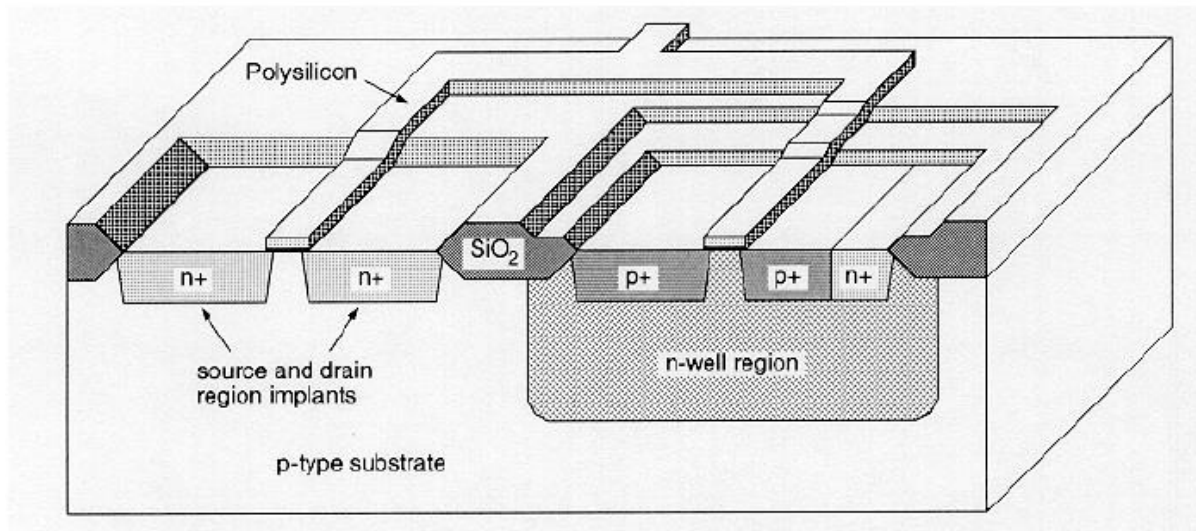
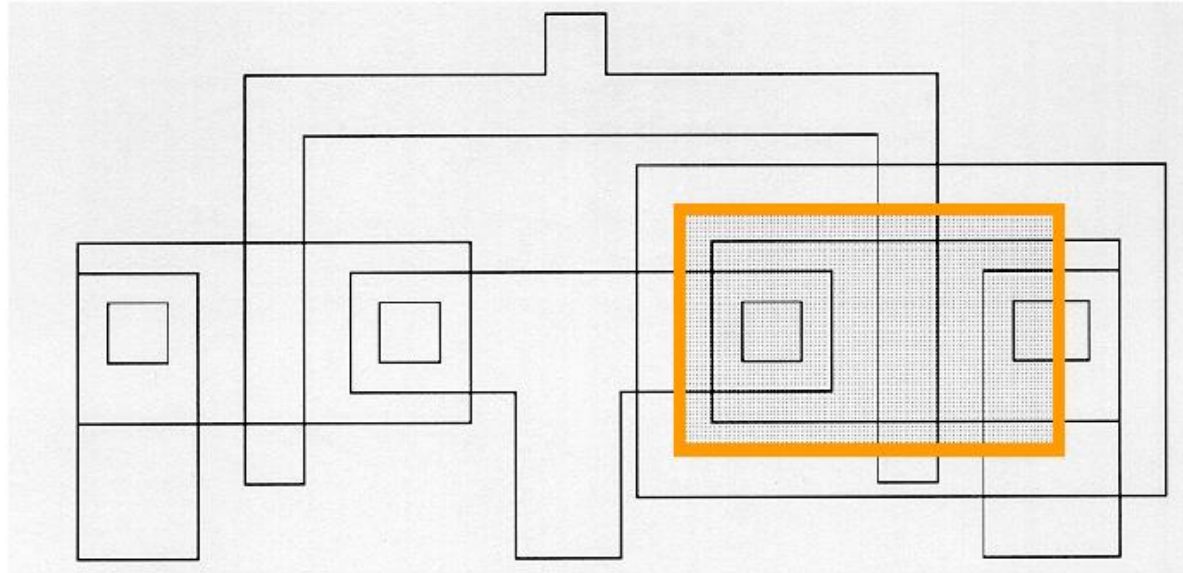
P-type substrate

- n-well region for PMOS
- thin gate oxide is grown on top of the active regions
- thick field oxide is grown in the areas surrounding the transistor active regions
- gate oxide thickness and quality affect the operational characteristics of the MOS transistor and reliability.



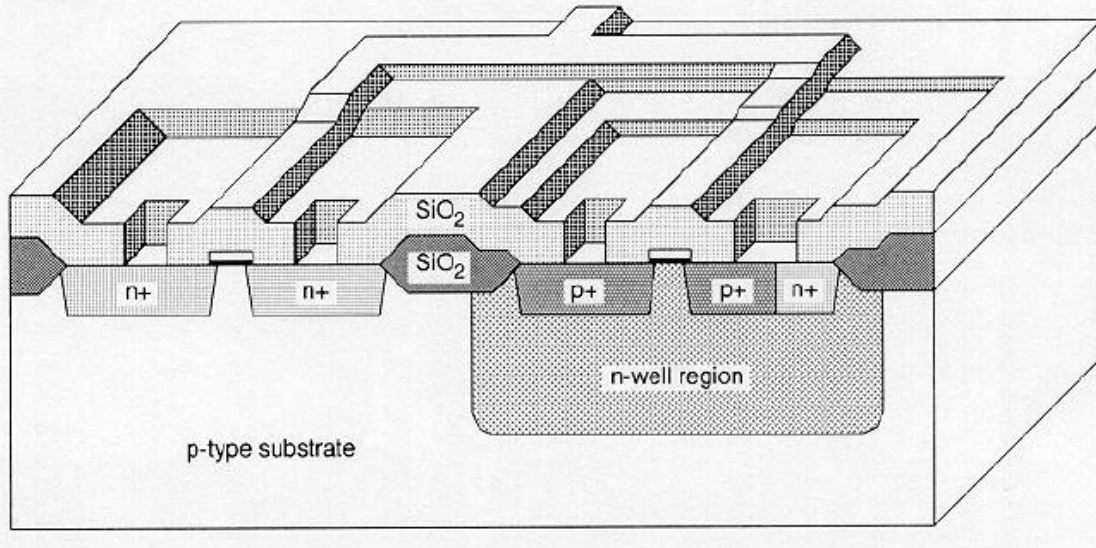
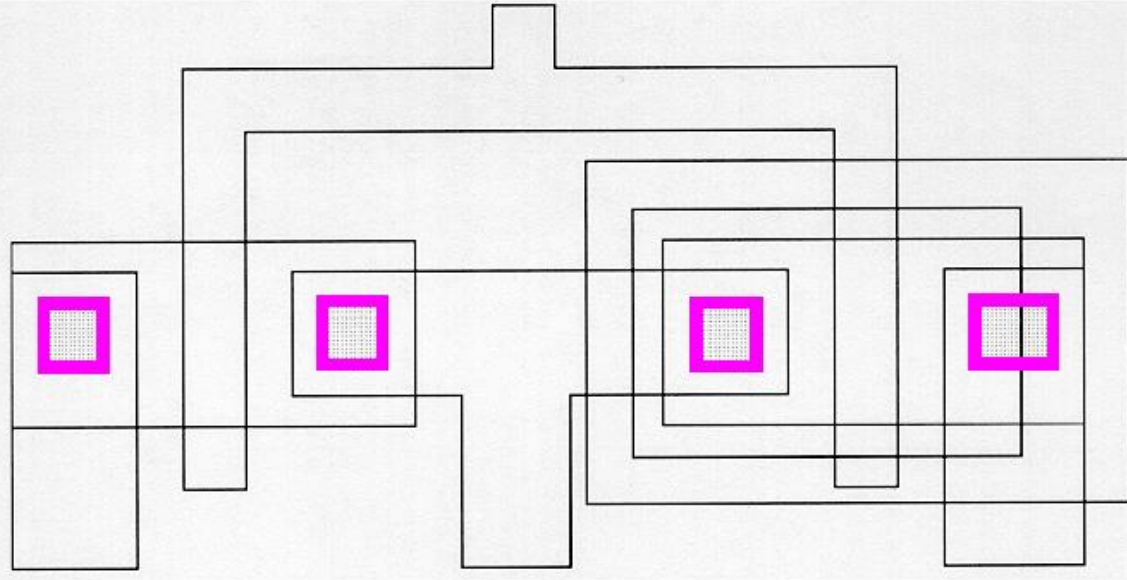
Poly

- ❖ The polysilicon layer is deposited using chemical vapor deposition (CVD) and patterned by dry (plasma) etching.
- ❖ The created polysilicon lines will function as the gate electrodes of the nMOS and the pMOS transistors and their interconnects.
- ❖ Also, the polysilicon gates act as self-aligned masks for the source and drain implantations that follow this step.



Implant

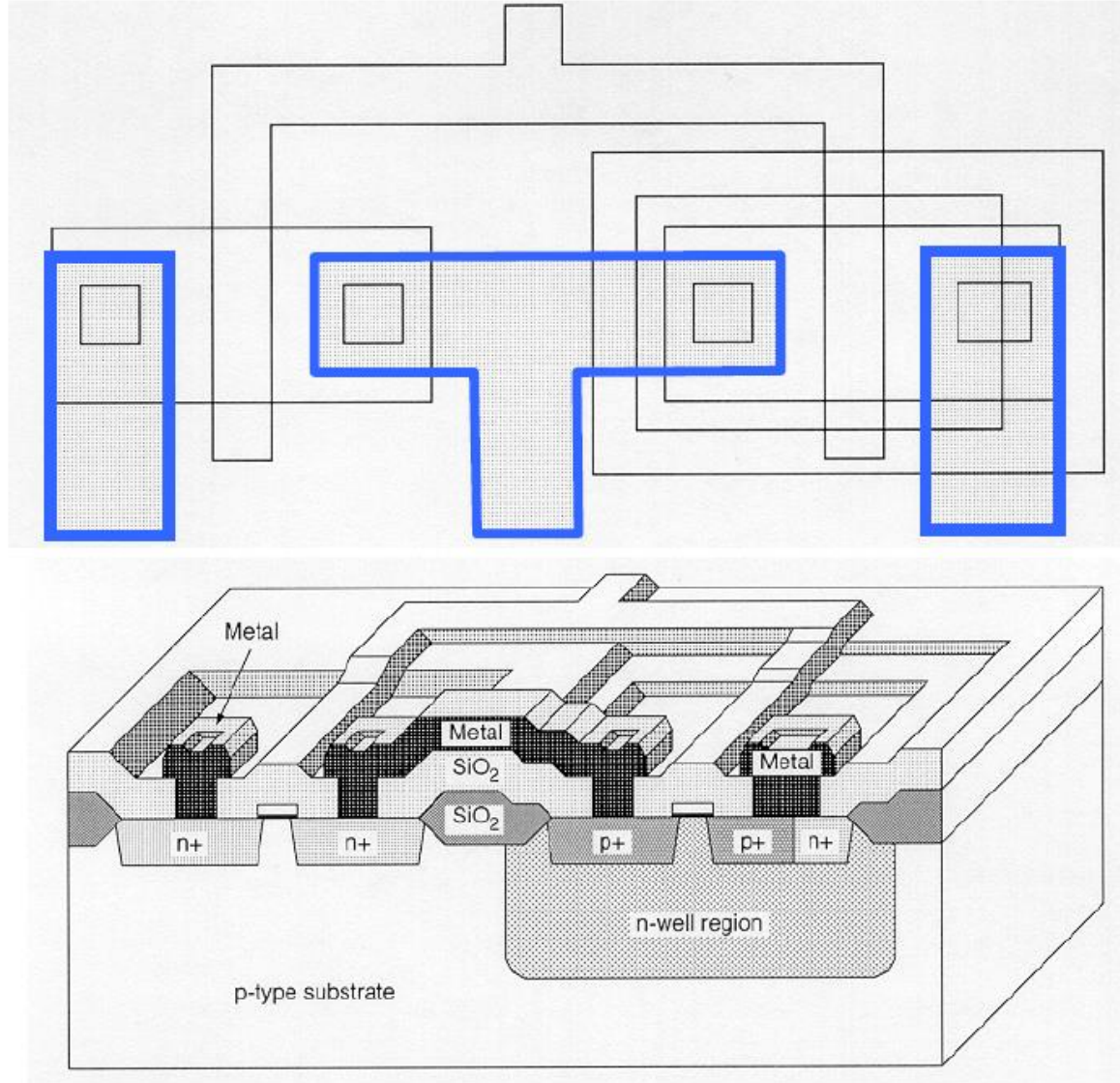
- ❖ Using a set of two masks, the n+ and p+ regions are implanted into the substrate and into the n-well, respectively.
- ❖ Also, the ohmic contacts to the substrate and to the n-well are implanted in this process step.



Contacts

- ❖ An insulating silicon dioxide layer is deposited over the entire wafer using CVD.
- ❖ Then, the contacts are defined and etched away to expose the silicon or polysilicon contact windows.
- ❖ These contact windows are necessary to complete the circuit interconnections using the metal layer, which is patterned in the next step.

metal

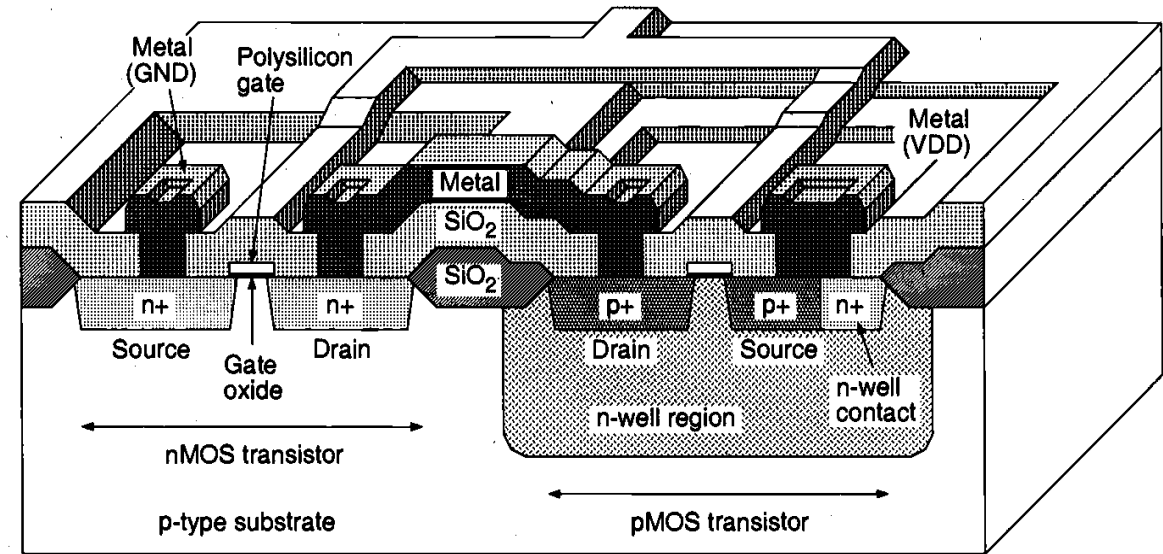
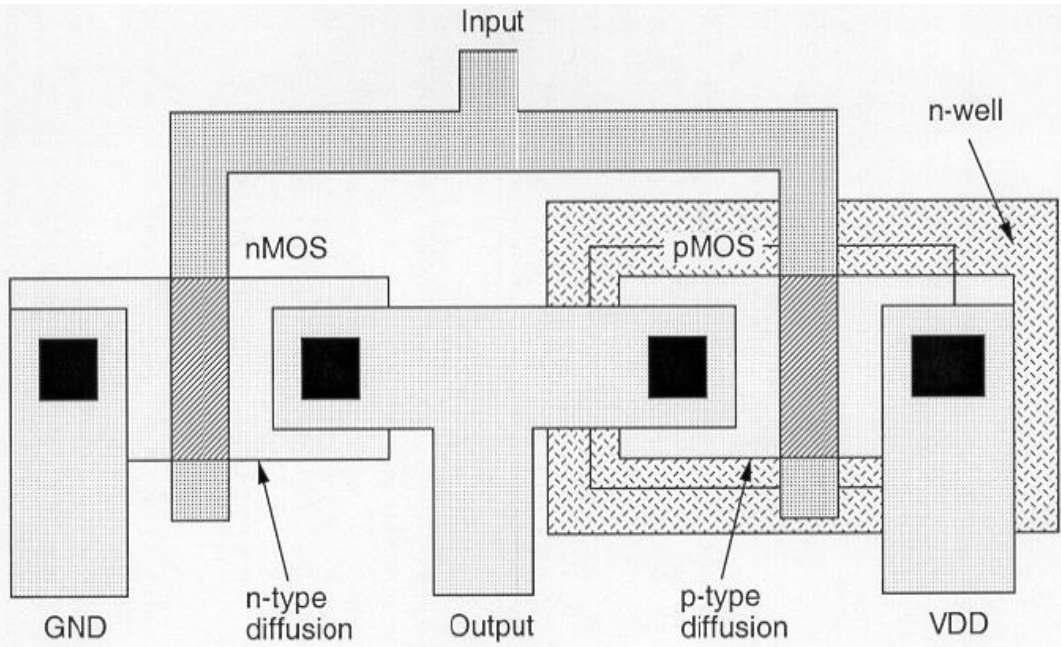


Metal

- ❖ Metal (aluminum) is deposited over the entire chip surface using metal evaporation, and the metal lines are patterned through etching.
- ❖ Since the wafer surface is non-planar, the quality and the integrity of the metal lines created in this step are very critical and are ultimately essential for circuit reliability.

Composite Mask Layout

- ❖ The composite layout and the resulting cross-sectional view of the chip, showing one nMOS and one pMOS transistor (in the n-well), and the polysilicon and metal interconnections.
- ❖ The final step is to deposit the passivation layer (for protection) over the chip, except over wire-bonding pad areas.





THANK YOU

Vinay Reddy

Department of Electronics and Communication

vinay@pes.edu

+91 9945730244